

Elemental Ratio Analysis of Lunar Surface Using X-Ray Fluorescence Spectroscopy

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Abstract

This paper reports on the finer analysis of data taken by the CLASS instrument aboard the Chandrayaan-2 spacecraft, based on X-ray fluorescence (XRF) spectroscopy measurements for elemental composition of the Moon's surface. The study adopted a new spatial interpolation method instead of traditional temporal averaging, thus improving the resolution and accuracy of the data. The use of data screening based on solar zenith angle segregation separates daytime and nighttime spectra to ensure effective removal of background noise. For the estimation of oxygen abundance, a new stoichiometric approach is applied, overcoming detection challenges generally inherent in low-energy X-rays. The study also involves area-weighted averaging at sub-pixel resolution so that the data can still be mapped accurately despite changes in data densities. These results were tested against Apollo, Luna, LP-GRS datasets and confirmed the presence of silicon, aluminum, magnesium, iron, calcium, sodium, titanium, manganese, and oxygen. The analysis reveals compositional heterogeneity and distinct geological regions, including Feldspathic Highland Terrane, mare basalts, and the Procellarum KREEP Terrane. Such insights are useful in understanding the geological evolution of the Moon and planning further missions for lunar exploration and resource utilization.

Key words: CLASS, X-ray fluorescence spectrometry, Spectrum, Background Spectrum, Geotail Passes, Line Flux Ratios, Compositional Groups, Meshgrid, Spatial Interpolation

1. Introduction

The elemental and mineral composition of the Moon's surface offers critical insights into its geochemical evolution and crust formation. Key elements such as magnesium, silicon, aluminum, calcium, iron, titanium, and sodium play a pivotal role in unraveling the Moon's geological history, aiding future exploration and identifying potential habitat regions.

The first set of global refractory elemental maps was obtained using the Lunar Prospector Gamma-Ray Spectrometer (LP-GRS) (Elphic et al. [1998]), with data binned at 5 degree equal area pixels. The estimation of elements such as magnesium, aluminum, and silicon presents significant challenges due to the instrument's

moderate spectral resolution and the high background noise in its detectors compared to the more precise measurements achievable with X-ray spectrometers. Despite these limitations, LP-GRS remains a critical tool for studying lunar composition at global scales.

In addition to orbital measurements, we also have access to comprehensive data from lunar soil samples returned by the Apollo and Luna missions, as well as the recent Chang'e missions (Meyer [2005], Haskin and Warren [1991], Che et al. [2021]). These samples provide invaluable ground-truth data, offering deeper insights into the compositional heterogeneity of specific lunar sites, which enhances our understanding of local and regional lunar geology.

X-ray fluorescence (XRF) spectroscopy is a well-established technique from which we can determine elemental abundances. Apollo 15 and 16 XRF experiments mapped the Mg/Si and Al/Si ratios for about 20% of the Moon (Adler et al. [1972]). The Chandrayaan-1 X-ray Spectrometer (C1XS) (Grande et al. [2009]) determined elemental abundances of about 5% of the Moon (Narendranath et al. [2011]) and elemental ratios (Mg/Si and Al/Si) in the western part of Oceanus Procellarum (Weider et al. [2012]).

Now we have the Chandrayaan 2's Large Area Soft X-ray Spectrometer (CLASS) instrument onboard the Chandrayaan-2 mission which has been providing the XRF data since 2019. This instrument is superior to the earlier C1XS instrument. The mapping attempt using the CLASS instrument was also done by (Narendranath et al. [2023a]), in which the relative elemental weight percentages were mapped. The elemental weight percentages were derived from algorithm *x2abundance* which used raw XRF photon counts.

However, it might be possible to map the elemental abundances without deriving weight percentages, by directly plotting line flux ratios. Silicon, owing to its relatively uniform distribution across the Moon, serves as an ideal reference element, minimizing systematic uncertainties Narendranath et al. [2023a]. Optimized flux ratios further enable the exploration of relative abundances and elemental correlations, offering valuable geochemical insights. This methodology also holds promise for analyzing challenging elements like oxygen, as noted in Narendranath et al. [2023a]. Further employing novel methods like mesh grids and spatial interpolation, we can achieve sub-pixel resolutions even better than what is achievable by the CLASS instrument.

2. Techniques Evaluated

2.1. X-Ray Fluorescence (XRF) Spectrometry

During a typical solar flare, a significant amount of X-rays are emitted from the sun which triggers XRF emission from rock-forming elements in airless planetary surfaces. Athiray et al. [2013] XRF refers to a process in which high-energy radiation excites atoms by ejecting electrons from their innermost orbitals. As the atom returns to a stable state, outer electrons fill the inner shells, resulting in the emission of X-ray fluorescence radiation. This entire process occurs without any contact or damage to the sample. The emitted radiation acts much like a unique fingerprint for each atom. The most important transitions that are used to identify elements are given names: an $L \rightarrow K$ transition is called $K\sigma$, an $M \rightarrow K$ transition is called $K\beta$, an $M \rightarrow L$ transition is called $L\alpha$, and so on. The fluorescence from copper, for instance, is distinct from that of zinc or any other element in the periodic table. This uniqueness makes XRF an especially straightforward and practical tool for elemental analysis, which is why it finds extensive use across industrial, research, and educational applications.

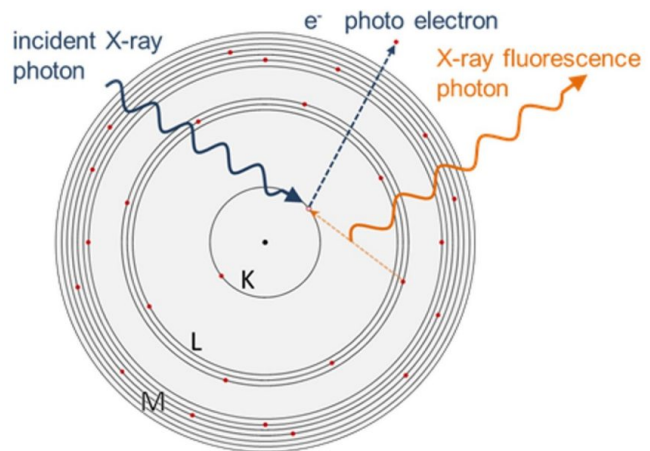


Fig. 1: Sketch of the X-ray fluorescence (XRF) process

2.2. CLASS Instrument

The Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS), developed by the ISRO Satellite Centre (ISAC), utilizes X-ray fluorescence spectra to analyze the elemental composition of the lunar surface. Magnesium, aluminum, silicon, calcium, titanium, iron, and sodium are among the primary elements that CLASS monitors in the Moon's X-ray fluorescence (XRF) spectrum. CLASS consists of an array of 16 swept charge devices (SCDs) operating in the energy range of 0.8–15 keV. Radhakrishna et al. [2020a]

These elements will be found using the XRF technique, which measures the distinctive X-rays that are released when they are activated by solar X-ray emission. The CLASS non-imaging spectrometer has gold-coated copper collimators, and each detector's field of view is 7 degrees by 7 degrees Full Width at Half Maximum (FWHM). This corresponds to a footprint of 12.5 km x 12.5 km at the spacecraft's 100 km altitude.

2.3. Geostationary Operational Environmental Satellite (GOES)

The Geostationary Operational Environmental Satellite System (GOES) is a series of geosynchronous satellites operated by the United States National Oceanic and Atmospheric Administration. The GOES system provides us with solar imagery, magnetometer data, solar X-ray data, and data on high-energy solar protons. GOES classifies solar flares based on the peak X-ray intensity measured in the 1-8 Å band. The classification system includes five classes: A, B, C, M, and X. Each class represents a tenfold increase in energy output, with A-class flares being the weakest and X-class flares being the strongest. For instance, an X2 flare is twice as intense as an X1 flare, and an X10 flare is ten times as intense as an X1 flare.

2.4. Data Processing and Uncertainty Estimation

Background averaging is performed using nighttime spectra. Statistical methods are applied to estimate

uncertainties in the flux measurements. A threshold of 1 standard deviation (1σ) is used to filter out noise.

3. Lunar Surface Chemistry

3.1. Minerals

Only four minerals – **plagioclase feldspar**, **pyroxene**, **olivine**, and **ilmenite** – account for 98 – 99% of the crystalline material of the lunar crust.

The exhaustive list is given below:

- **Pyroxene** – A group of magnesium-iron-calcium silicates, common on the Earth and Moon.
- **Clinopyroxene** – A form of pyroxene; typically contains some calcium; most common in mare basalts [$Ca(Mg, Fe)Si_2O_6$].
- **orthopyroxene** – A form of pyroxene; contains little calcium; most common in highland rocks [$(Mg, Fe)SiO_3$].
- **olivine** – A magnesium-iron(II) silicate; common on Earth and Moon [$(Mg, Fe)_2SiO_4$].
- **ilmenite** – An iron(II)-titanium oxide; more common in lunar basalts than in terrestrial basalts [$FeTiO_3$].
- **feldspar** – A group of alumino-silicate minerals; common in the crusts of the Earth and Moon.
- **plagioclase** – A form of feldspar; a calcium-sodium alumino-silicate [$(CaAl, NaSi)AlSi_2O_8$].
- **Anorthite** – A mineral; the calcium-rich extreme of the plagioclase feldspar; the most common mineral of the lunar crust, but not so common on Earth.
- **Anorthosite** – A rock consisting mainly of anorthite.

3.2. Compositional Heterogeneity

Once we get our relative elemental abundances, we can produce ratios, with some abundant elements like **Silicon** as reference. Then we plot the relative abundance, concerning silicon. Once we plot the data, we can see patterns emerging in the plot, with dots corresponding to certain elemental ratios getting clubbed together in some regions. These regions of higher density of data can be named compositional groups, indicating their unique mineral composition and geographical region on the moon. The major Highland Terrane

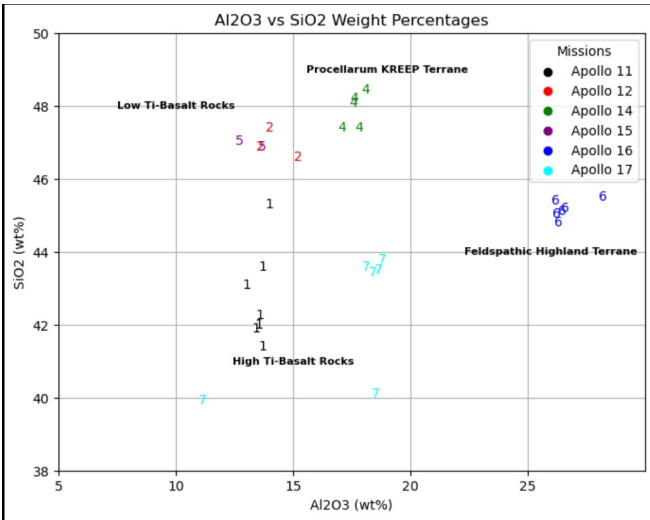


Fig. 2: SiO₂ vs Al₂O₃ weight percentage, generated from Lunar Soil Compendium (see appendix)

3.2.2. Mare Basalts

The composition of lunar mare basalts varies widely, depending on specific **mare regions** (darker regions of the lunar surface) and basalt types. They are generally high in FeO and TiO₂ compared to the Feldspathic Highlands Terrane (FHT) and display distinct elemental ratios due to their volcanic origin from the Moon’s mantle. The titanium can vary widely in these regions, and hence this group may further be divided into **High Ti** and **Low Ti Basalts**.

3.2.1. Feldspathic Highland Terrane

The Feldspathic Highlands Terrane (FHT) is a region that spans a significant portion of the lunar far side. It is characterized by a particularly high concentration of feldspathic minerals, which are rich in aluminum, with plagioclase feldspar being the most prevalent. This distinctive mineral composition sets the FHT apart from other lunar regions and offers valuable insights into the early history and geological processes of the Moon. The terrain’s formation processes provide essential clues about the Moon’s crustal evolution and the impacts that shaped its surface.

The elemental profile shows high Al₂O₃ and CaO relative to SiO₂, indicative of its anorthositic, plagioclase-rich crust and low levels of FeO, MgO, TiO₂, Na₂O, Cr₂O₃, and MnO.

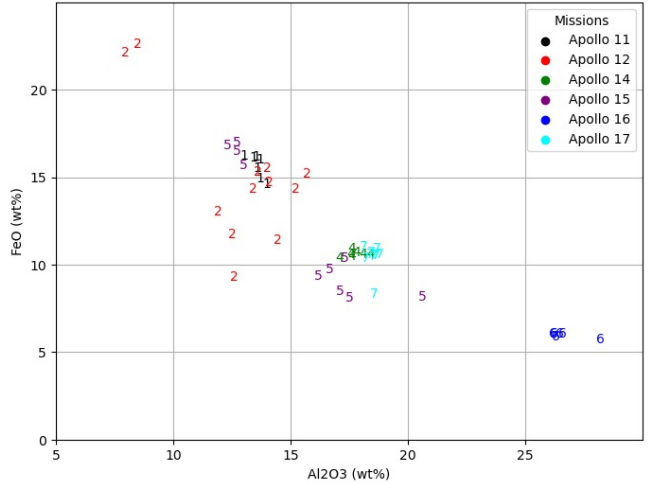


Fig. 3: FeO vs Al₂O₃ weight percentage, generated from Lunar Soil Compendium (see appendix)

3.2.3. Procellarum KREEP Terrane

The Procellarum KREEP Terrane (PKT) on the Moon's near side is characterized by high concentrations of incompatible elements, including potassium (K), rare earth elements (REEs), and phosphorus (P), collectively referred to as KREEP. This region may also contain other **exotic elements** like radioactive thorium deposits and is certainly a key region of interest.

3.2.4. Olivine-Bearing Rocks

It is a minor compositional group but an interesting one. Olivine-bearing rocks contain a higher concentration of **magnesium (Mg)** and **iron (Fe)**, with olivine (along with pyroxene) as the dominant mineral phase.

4. Mapping the ratios onto a lunar base map

The process begins by collecting high-quality data from the CLASS instrument on Chandrayaan-2, which measures elemental abundances on the lunar surface. The line flux data obtained is now used to map it onto a lunar base map. To map it, we calculate the line flux ratios of an element by dividing its flux value by the flux values of other elements. The elemental ratios of interest, such as Mg/Si; and Al/Si are thus calculated.

Once these ratios are calculated, they are then overlaid onto a lunar albedo map. This approach allows us to identify regions rich in certain elements.

5. The case of oxygen

Oxygen is available on the lunar surface in almost uniform quantity, but it is difficult to measure the abundance using the XRF methods. It is because as we know XRF primarily works by exciting atoms to release fluorescent X-rays when bombarded with high-energy X-rays (for example solar flares), but since Oxygen has a relatively low atomic number ($Z = 8$), it emits X-rays at very low energy (~ 0.52 keV), which is often below the detection limit of XRF equipment. The CLASS instrument is capable

of measuring from an energy range of 0.8 KeV to 15 KeV. Radhakrishna et al. [2020b] Therefore we use Lunar Prospector Gamma Ray Spectrometer (LP-GRS) abundance values for O, Ca, and Ti (except for spectra where Ca and Ti lines are observed in CLASS data). Narendranath et al. [2023b]

6. Sub-pixel resolution maps

To map the lunar surface with greater detail and accuracy, we initiated the process by constructing a mesh model of the Moon. The foundation of this model is a quad sphere. To create a quad sphere from a standard sphere, we can subdivide the sphere into quads by using a subdivision algorithm, the icosphere method. Through this iterative process, the resulting structure evolves into a highly uniform grid of interconnected quads, evenly distributed across the sphere's surface. By subdividing the sphere into finer quads, we effectively create a grid of smaller units on the surface. These units can be considered sub-pixels, as each unit represents a small section of the lunar surface. This enhancement of the resolution of elemental ratio datasets is crucial for capturing localized variations and providing detailed insights into regional patterns. After creating a mesh of uniform area over the moon, we calculate the values for the cells using weighted spatial aggregation instead of temporally adding them. The approach ensures that raw data contributions to each cell are proportional to their overlap percentages. In employing this method, we assume that we have dense raw data coverage across most of the lunar surface, except for a few sparse areas where limited contributions may increase variability. The weighted aggregation was preferred over simple averaging to preserve spatial fidelity and minimize potential biases caused by uneven data distribution. We eliminate outlier pixels, such as those with zero values or extreme deviations from the expected mean, by clipping the value range. This results in better statistics, ensuring

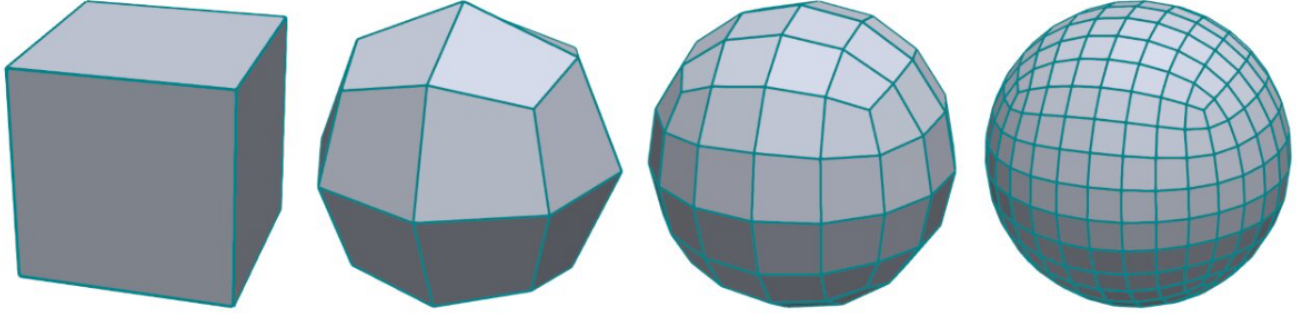


Fig. 4: Concept mesh creation of a sphere. By reducing the mesh cell size, we can improve the resolution.

more accurate and reliable line flux ratios for the mesh-grid pixels

The methodology bears similarities to convolution processes commonly used in image processing, where local neighborhood data is aggregated using predefined weights. Here, the overlap percentages function as a dynamic kernel, adapting weights based on the spatial relationship of raw data to the grid cells. Thus, we use spatially neighboring data to improve the data for a particular cell, commonly referred to as interpolation.

6.1. Creating mesh

The algorithm to generate a quad sphere iteratively subdivides an initial hexahedron. We first process a CSV file containing grid center coordinates (latitude and longitude) and optional elemental composition ratios, representing locations on the Moon's surface. It calculates the bounds of quadrilateral grid cells around each center point to create a structured mesh. Each quadrilateral's vertices (four corners) are calculated by adjusting the center coordinates up and down in latitude and longitude, accounting for spherical distortion using trigonometric functions.

The area of each pixel in the new mesh can be controlled. We can improve the spatial resolution from $12.5 \text{ km} \times 12.5 \text{ km}$ to better resolutions. As we improve the spatial resolutions, the data from scientific instruments, such as the CLASS,

are mapped across more quads, with each quad representing a smaller area of the Moon's surface. Consequently, the same data points may overlap multiple sub-pixels, which enhances the resolution of the analysis. By distributing the instrument's readings over a greater number of smaller units, we can achieve a more detailed representation of the lunar surface features and variations.

7. Methodology

A CLASS instrument onboard the Chandrayaan 2 orbiter is continuously operated on the day side where the Solar X-rays fall upon the lunar surface and excite the atoms which then emit their characteristic X-rays, which are picked up by the CLASS instrument. There are sixteen X-ray detectors called Swept Charged Detectors (SCDs) (Pillai et al. [2021]) to detect the X-ray photons as time-tagged events. Since the instantaneous field of view of the mechanical collimators on CLASS defines a ground footprint of $12.5 \text{ km} \times 12.5 \text{ km}$ at the 100 km altitude of the Chandrayaan-2 orbiter, the lowest time bin is 8s (time taken to sweep 12.5 km with ground trace velocity of 1.56 km/s). The sixteen spectra are added to get the combined spectra. The combined raw XRF data is an L1 data product in FITS format from which we will begin the work.

7.1. Timing the Solar Flare Events

The solar X-ray flare events involve with spike in X-ray intensity which is favorable to get stronger emission lines from the elements. To time such events, we use the GOES X-ray flux data. In the metadata, we can collate the timings and class of flare events and then in upcoming data processing, use them to refine our results.

The solar X-ray flare events are classified into various classes (A, B, C, M, X) based on increasing X-ray intensity recorded. Different elements respond differently to a given solar flare class. It is observed that lighter elements like Mg, Si, Na, or Al are excited in low-intensity flare classes (like B or M) and heavier elements like Fe, Ca or Ti require more intense flare events like M and X (Hall [2023]). Consequently, precise timing of X-ray events is essential for detecting emission lines of heavier elements.

7.2. XRF Spectral Screening

Each FITS file is associated with a pixel of $12.5 \times 12.5 \text{ km}^2$ with an integration time of 8 seconds, on the lunar surface. We need to segregate the files into day and night time. We do this by getting the solar zenith angle data from the CLASS instrument. Files having solar zenith angle of $> 90^\circ$ are associated with nighttime data and the rest is daytime data.

This segregation was crucial as nighttime data is used for constructing the **background X-ray spectrum**. The background x-ray spectrum consists mostly of cosmic rays, natural radioactivity, and solar wind (Okada et al.). The background spectrum is crucial since these random emissions constitute noise in the observed data. Incorporating these emissions into a well-defined background spectrum enables effective noise reduction during subsequent analyses (Narendranath et al. [2023a]). However, the background spectrum achieves an unusual spike around the energy channel corresponding to the Al K- α line. This occurs during **geotail passes** of the moon's orbit around

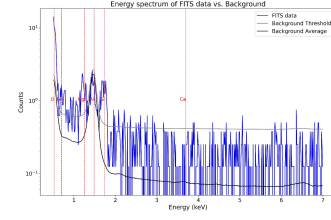


Fig. 5: This figure shows counts vs energy graph without geotail passes removal

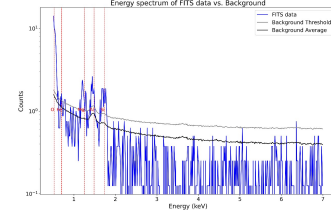


Fig. 6: This figure shows counts vs energy graph with geotail passes removal

the sun when the moon enters the bow shock region created by Earth's magnetosphere and solar wind. These cause particle interactions and spiking in the background spectrum. (Narendranath et al. [2023a], Narendranath et al. [2014]).

We remove such files by identifying such unusual peaks. We do this by constructing a mean line of that spectrum, and checking if any unusual spikes lie above 1σ (standard deviations) from the mean line. If such peaks are detected, we discard those files for further calculations.

To ensure temporal consistency, we performed this process every month. Specifically, nighttime data collected over one month was averaged to generate a representative background spectrum for that period. This monthly averaged spectrum was then employed in subsequent spectral analyses to enhance data quality and reliability.

7.3. Line Flux Calculations

The emission lines are extracted from the daytime files. From each daytime associated file, an XRF plot is constructed. Each XRF plot represents photon counts as a function of energy, providing a detailed energy spectrum. The continuous energy

spectra are binned into 2048 discrete energy channels (Narendranath [2010]). The incoming photons are then chunked into these energy channels to get the plot. To account for background noise, the corresponding monthly background X-ray spectrum is superimposed on each file's XRF plot, enabling a clearer distinction between the emission lines and randomized background contributions Narendranath et al. [2023a].

The excitation energies of the elements we need to investigate, than can be compared to the corresponding energy channel and the line (photon count). A key challenge arises when the photon count (line intensity) associated with an element is too weak to be reliably identified. To address this, a mean line of the averaged background radiation is constructed. A threshold is then defined as one standard deviation (1σ) above the mean background level. Peaks that do not exceed this threshold are considered statistically insignificant and are discarded.

After the filtering process, the analysis focuses on the actual peaks identified in the XRF spectra. Although we need the line flux data, we have photon counts in the y-axis. This idea is discussed in (Narendranath [2010], Margu   et al. [2009]) that the photon count rate is the convolution of line flux and the spectral response of the detector. Practically, this proportionality allows us to treat the count rate as an equivalent representation of the line flux, particularly when calculating line flux ratios. Thus, we replace the count rate with its corresponding line flux value and compute the desired line flux ratios.

7.4. Mapping

Once we get the line flux ratios for each FITS file, we create a monthly catalog in CSV file format. It will contain the line flux ratios and the geographical location of the pixel which we got from the FITS file's metadata. The catalog is provisionally mapped onto the lunar surface using

LROC. However, this initial mapping contains overlapping of pixels and at best $12.5 \times 12.5 \text{ km}^2$ resolution.

To achieve sub-pixel resolutions, we employ the idea of mesh grids. We divided the lunar spherical surface into desired surface pixels (preferably better than the resolution previously mentioned). We superimpose this new mesh grid onto our previously plotted FITS pixels onto the lunar surface. Now for each new mesh-grid pixel, we will have various FITS file pixels overlapping with it. We took a weighted average of the data, depending upon the area overlapped with the new mesh-grid pixel for the FITS file pixel.

Now we will have new line flux ratios for each new mesh-grid pixel which is the weighted average of all the FITS file pixels intersecting that region. With this, we can solve the overlapping issue, and also achieve sub-pixel resolutions.

7.5. Validation

Multiple attempts have already been made to map the elemental abundances of the lunar surface using XRF spectroscopy. Apollo 15 and 16 XRF experiments mapped the Mg/Si and Al/Si ratios for about 20% of the Moon (Adler et al. [1972], Andre et al. [1977]). We also have the L2 data of elemental abundances from Chandrayaan-2. Further, we have the Rock Soil data from the various lunar missions so far. We can cross-check our results from these data for validation. Narendranath et al. [2024]

7.6. Compositional Groups

The vast swathes of the lunar surface can be divided into various compositional groups depending upon relative elemental abundance or specific mineralogy. The crucial groups include **Feldspathic Highland Terrane** which contains a relatively high amount of Aluminium, **Mare Basalts** which contain relatively high Magnesium, and **Oceanus Procellarum KREEP Terrane**, which contains a high amount of trace elements like Thorium, Phosphorous, and Potassium. The

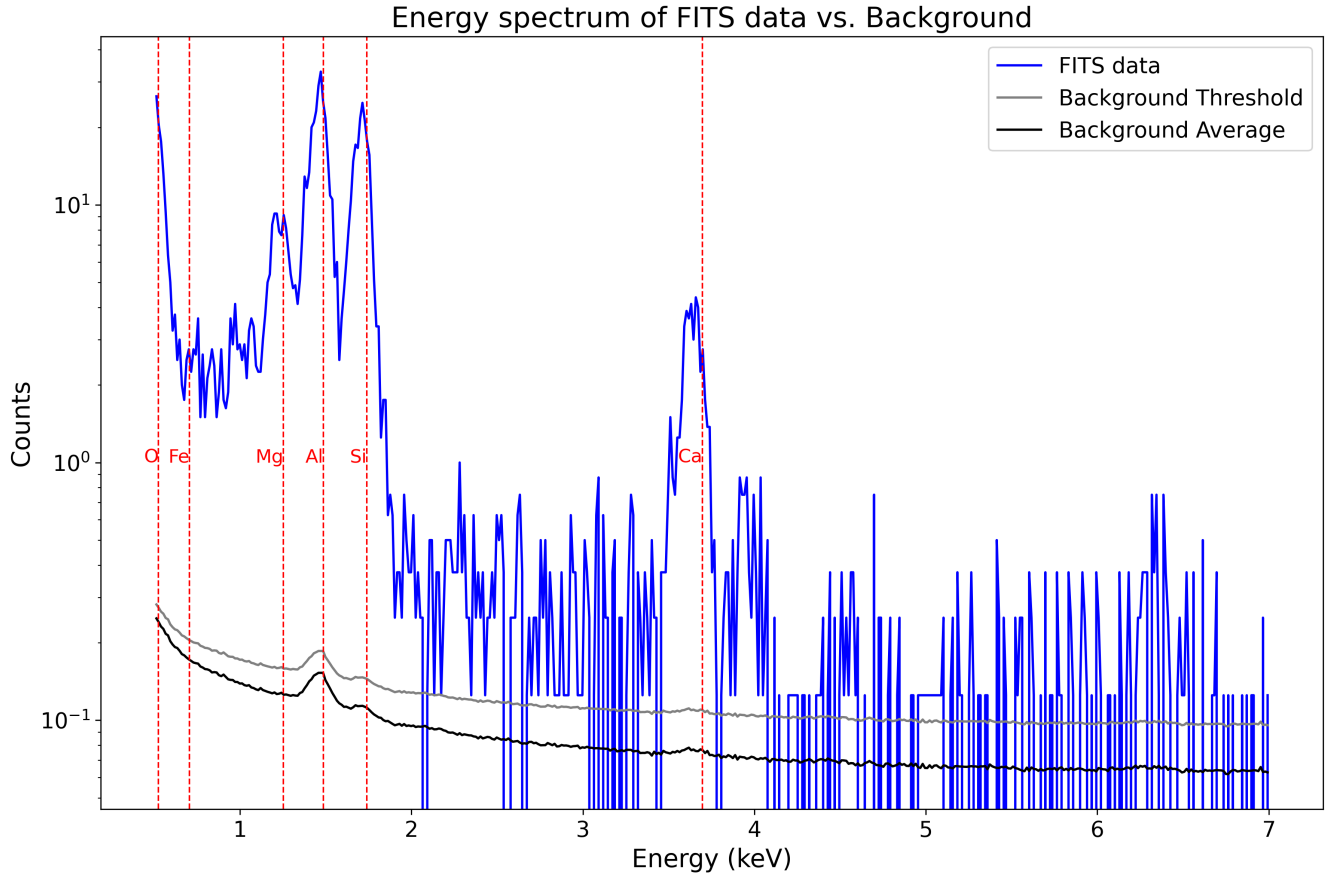


Fig. 7: Sample XRF spectrum with background data and threshold. We can clearly see XRF lines respectively for O, Fe, Mg, Al, Si, Ca

elemental ratios (Mg/Si and Al/Si) in the western part of Oceanus Procellarum have been mapped by (Weider et al. [2012]). The compositional groups can be observed by plotting the raw line fluxes obtained for each element and plotting them against some reference (preferably data for silicon). We will see clustering of data or a high Standard deviation in specific ranges of elements, which constitute a specific compositional group or compositional heterogeneity. (Narendranath et al. [2023a], Gloudemans, A. J. et al. [2021]).

8. Result And Discussion

8.1. XRF plots

After getting our daytime and monthly average background spectrum, we plotted the XRF plots of counts vs energy. Note that discrete binned energy channels were converted to corresponding true

energy by conversion formula. From the literature data, we marked the excitation energies (K- α lines) for corresponding elements. We also plotted the background spectrum and calculated our threshold count to accept the peaks (1σ). In these plots, we are obtaining positive indications of the presence of Mg, Si, Al, Ca, Fe, and even oxygen. However such high counts for oxygen could be due to its low threshold excitation energy and secondary effects like matrix effects. The same is observed in certain regions for Iron due to systematic uncertainties Narendranath et al. [2023a]

8.2. Aluminium and Magnesium Results

As we discussed in methodology, the line fluxes are equivalent to photon counts (Margu  et al. [2009]), we can use the count's value to calculate the Al/Si line flux ratios, for the entire XRF catalog. The

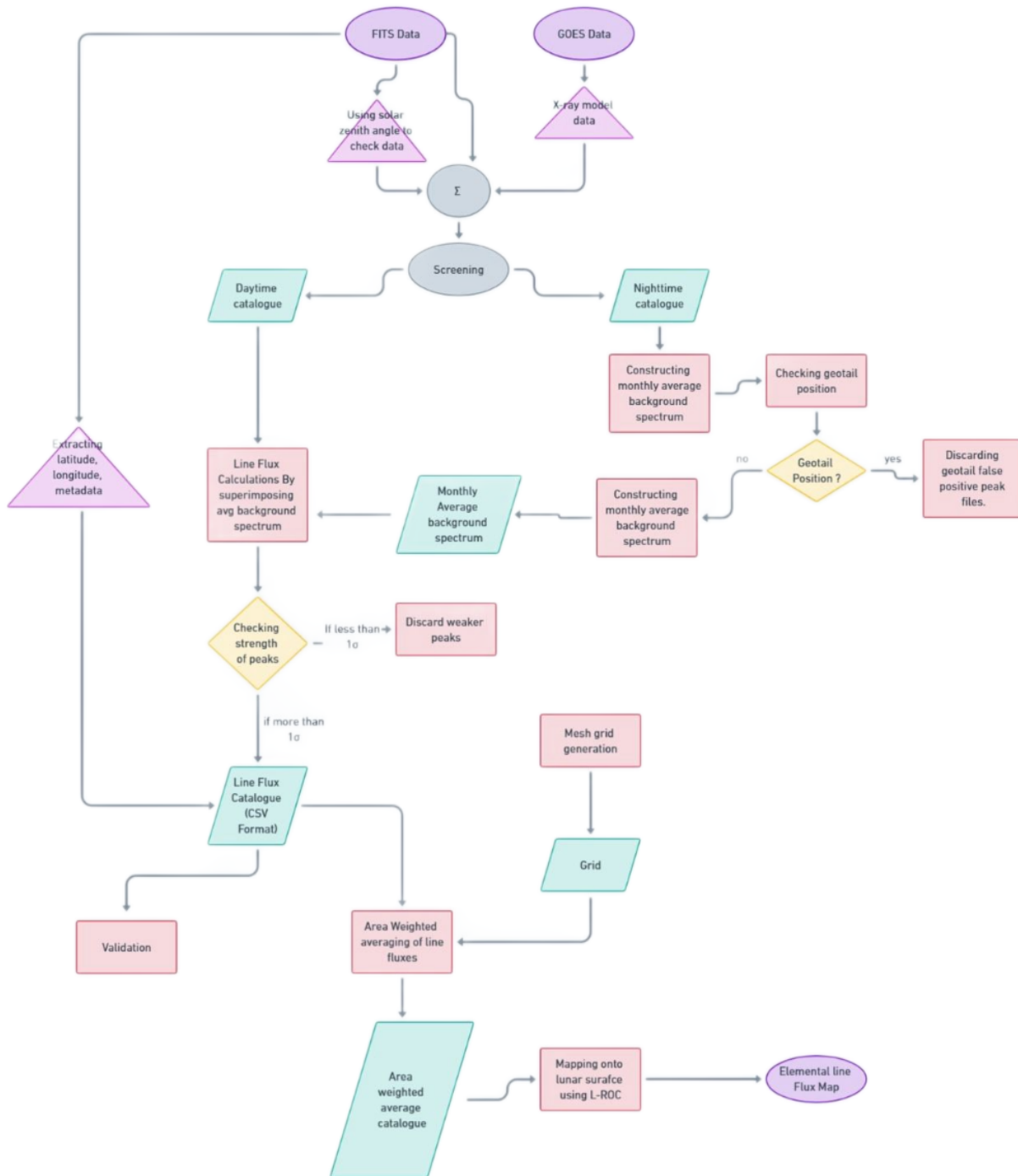


Fig. 8: Flowchart of Methodological process. The pink triangles depict extraction of data from depository, green parallelograms depict data catalogues, Red boxes depict some processes, yellow rhombuses depict decisions.

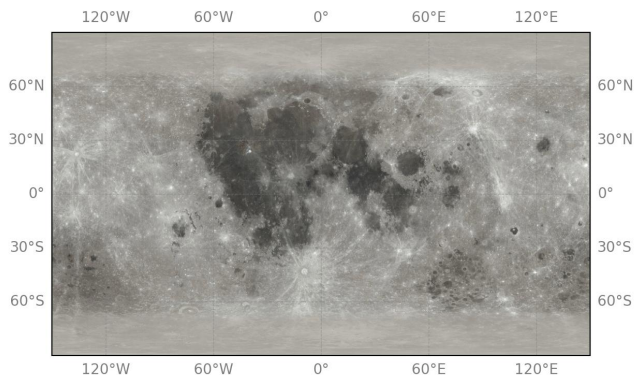


Fig. 9: Sample Equidistant Cylindrical Albedo Lunar Map ($300^\circ \times 180^\circ$) used for plotting Ratios.

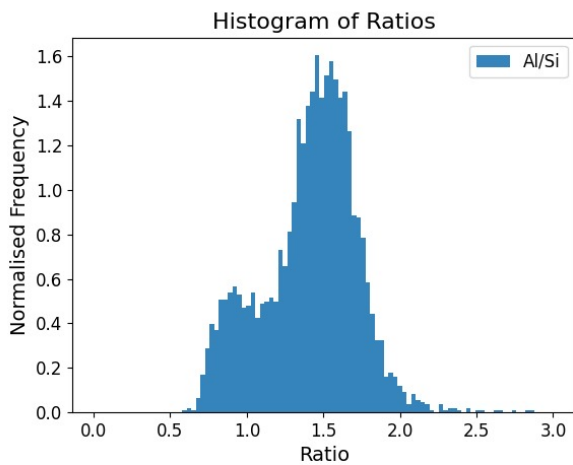


Fig. 10: Histogram showing Normalized frequencies of the Al/Si ratios taken at 100 km resolution

ratios that we get are then plotted onto a histogram as shown. For Al/Si we plotted two histograms on two different resolutions, and the results are contrasting. So we only plot the rest at 12.5 km resolution for better accuracy.

The combined ratio plots for Al/Si and Mg/Si are shown below. A contrasting feature we can see is that the Al/Si plot fits a bimodal Gaussian distribution, while a unimodal Gaussian distribution fits Mg/Si. The bimodal Gaussian predicts two distinct populations of ratios (which could be two geographical regions/minerals contribution) This is further validated by ratio clustering done on the Al/Si distribution, which neatly divides the ratios into two clusters or

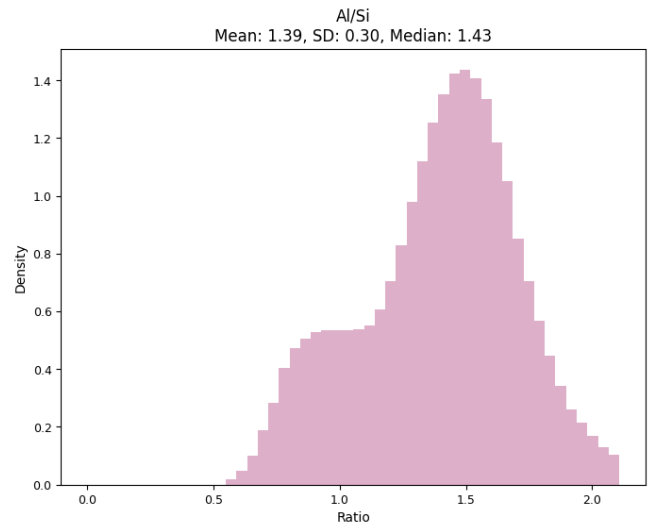


Fig. 11: Histogram showing Normalized frequencies of the Al/Si ratios taken at 12.5 km resolution. This histogram has a bimodal Gaussian distribution.

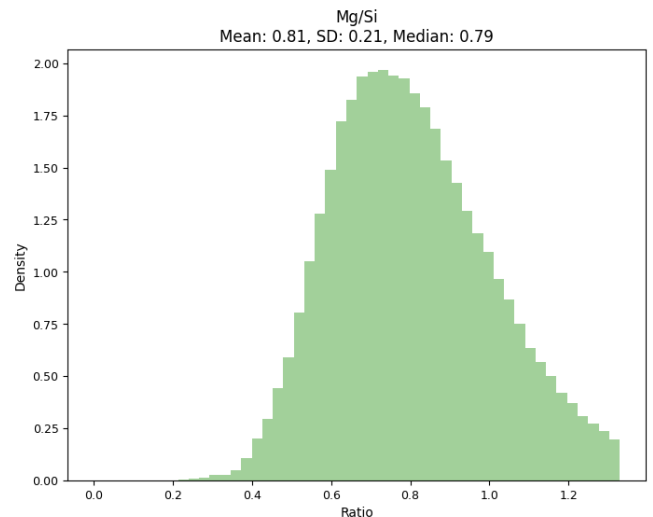


Fig. 12: Histogram showing the normalized frequencies of Mg/Si ratios at 12.5 km resolution. It has a unimodal Gaussian distribution.

compositional groups. Consequently, the literature data describes the regions having high Al/Si ratios as belonging to Feldspathic Highland Terranes (the lighter parts of the lunar surface), while lower ratios belong to mare basaltic regions (darker regions).

The Mg/Si has an unimodal Gaussian distribution, which suggests a less apparent distinction between the geological features in the Mg/Si map, which

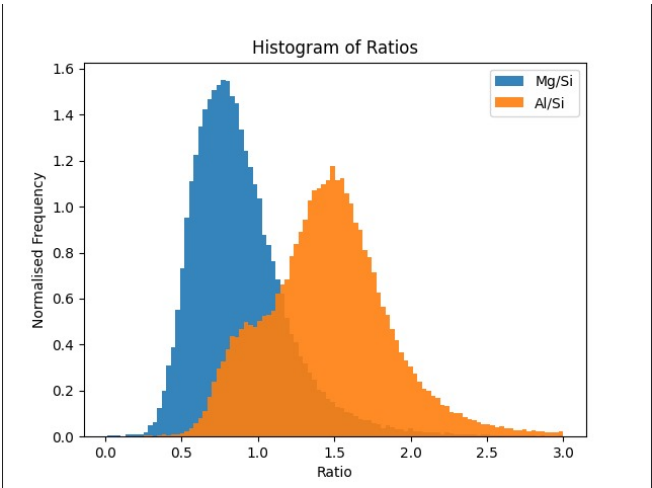


Fig. 13: Histogram plot of normalized frequencies at both Mg/Si and Al/Si ratios at 12.5 km resolution. We see an inverse relationship between the Mg/Si and Al/Si ratios. The bimodal distribution of Al/Si constitutes two geographical regions, with lesser ratios belonging to Lunar Maria and higher ratios belonging to Highland Terranes. The Mg/Si ratio on the other hand has a single Gaussian distribution, and has lesser geographical distinction, although has a relatively higher presence in Lunar Maria

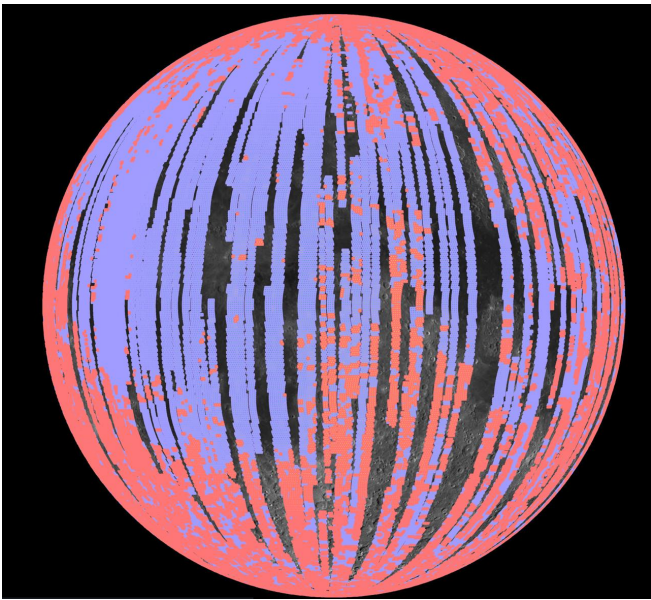


Fig. 15: Al/Si ratios plotted onto the lunar surface. The blue regions denote the compositional/cluster group (mares) having lesser ratio values. The red regions denote the compositional group (Highland Terrane) having higher ratio values.

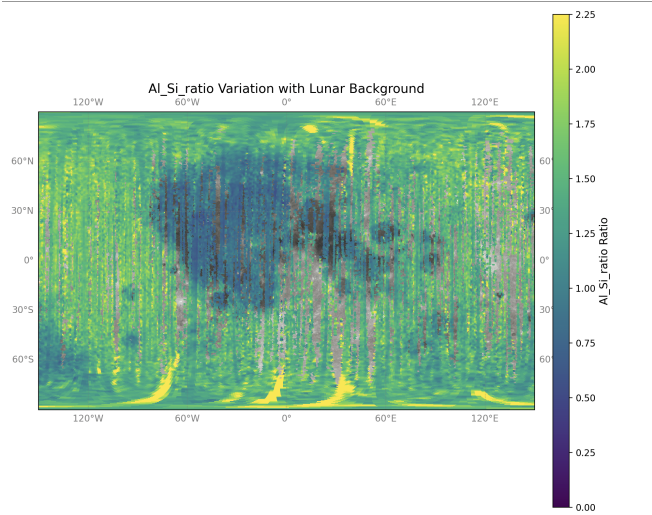


Fig. 14: Equidistant Cylindrical plot at 12.5km resolution, Al/Si ratios for nearside of moon. We can clearly distinguish high Al in Highland Terrane and low Al in Mare Basalts.

is also discussed in Gloudemans, A. J. et al. [2021]. The literature also describes that Lunar Maria contains a higher frequency of ratios of Mg/Si compared to Al/Si, hence showing an inverse relationship between Al/Si and Mg/Si intensity ratios (Al-rich Highlands and Mg-rich Maria).

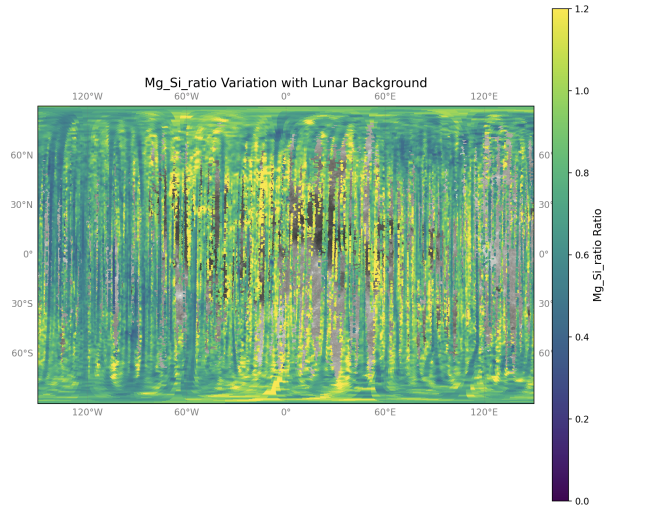


Fig. 16: Equidistant Cylindrical plot for Mg/Si ratios at 12.5 Km resolution for the nearside of moon. We can clearly see the high Mg presence in Mare Basalts and relatively low presence in Highland Terranes.

The comparatively higher Al/Si intensity ratio values in the highlands are due to the presence of anorthosite type of rocks (mostly plagioclase feldspar). The mare regions, however, are characterized by high Mg/Si intensity ratio values compared to the highland terrane regions due to

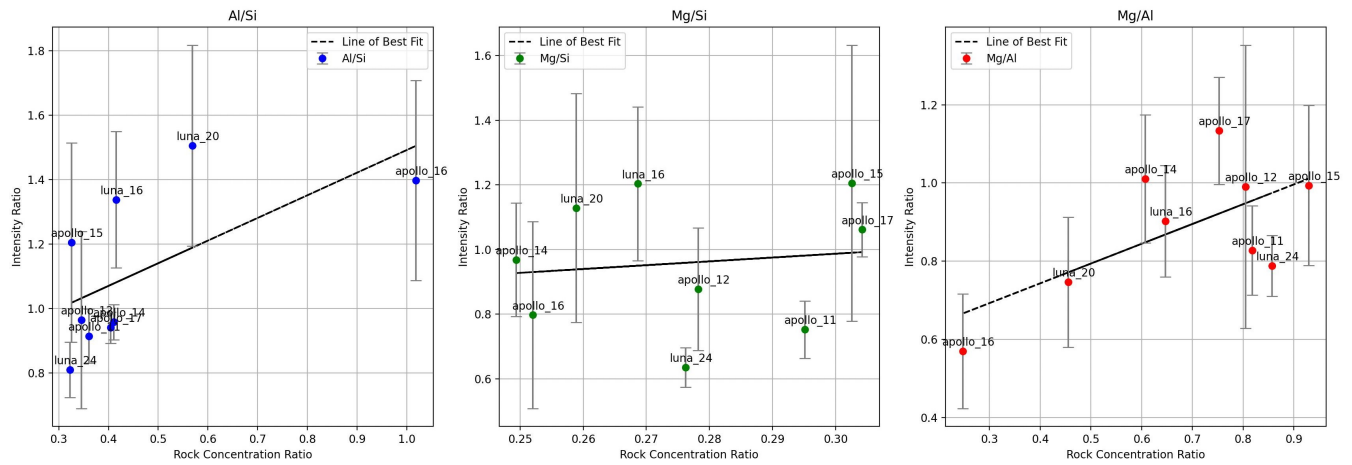


Fig. 17: Intensity ratio as a function of the measured Lunar rock concentration from various missions; of Al/Si (left), Mg/Si (middle), and Mg/Al (right) with the black line showing the optimal linear fits. This has a positive correlation with the results in Gloudemans, A. J. et al. [2021]

relatively Mg-rich basaltic lavas, typically abundant in the mineral pyroxene.

In the histogram plots, the SD and mean for Al/Si plots are high. This predicts the dominant presence of Al over Si and also has compositional heterogeneity. On the other hand Mg/Si has lesser SD and mean, suggesting a more uniform and lesser presence on the lunar surface.

We plotted the Al/Si ratios on the lunar surface as shown in Fig. 14. The darker red regions contribute to our Feldspathic Highland Terranes, which contain relatively high amounts of Aluminium or value of Al/Si ratios. Similar plots had been done in Narendranath et al. [2023a], and it shows a positive correlation for the Al wt map.

Further cross-validation is done by using the rock samples from the Luna and Apollo missions. The lunar sample compendium, Meyer [2005] provides exhaustive data on the elemental concentration of the rock samples from these missions. Such concentration ratios from lunar rock samples are compared to the average XRF intensity ratios. These values have been plotted in Fig. ??, and are used to derive a linear relation between the measured intensity and concentration ratio. The

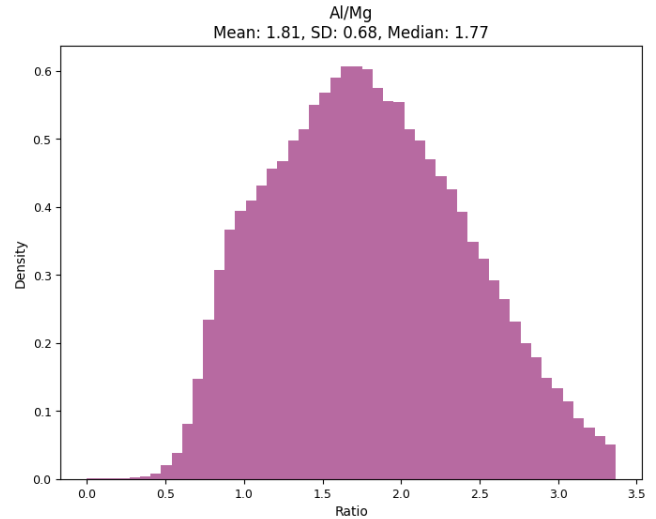


Fig. 18: Histogram showing normalized frequencies of Al/Mg

results again have a positive correlation with the results in Gloudemans, A. J. et al. [2021].

Further, we can go one step to plot the Al/Mg ratios, which again yield us a bimodal Gaussian distribution. The distribution has a high standard deviation, which is predictable since both elemental ratios are concentrated in two distinct regions. The lesser valued ratios belong to Mares while higher valued ratios belong to Highland Terranes. The ratios have values close to 1 and can represent the boundary or transition zones of Mares and

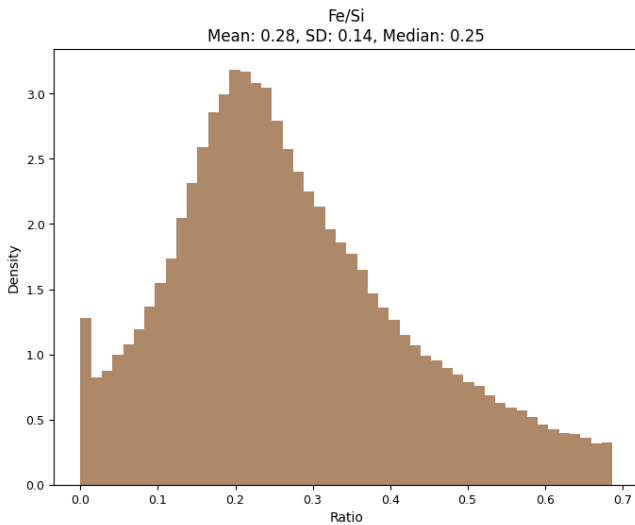


Fig. 19: Histogram plot of normalized frequencies of Fe/Si ratios. A relatively average SD suggests compositional uniformity in comparison to Al/Si ratios.

Highland Terranes. So Al/Mg ratio can provide us with this information.

8.3. Iron and Titanium

The iron has a comparatively lesser presence on the lunar surface, as indicated by a lesser mean value, relative to silicon. As per literature, (Narendranath et al. [2023a], Haskin and Warren [1991]), iron is predominantly present in Mare basalts mostly in the form of minerals like olivines, iron oxides, and ilmenites. The Fe presence in the highlands is pretty limited, and there are instances of small regions possessing unusually high Fe presence in the Highlands, which has been argued to be due to systematic uncertainties as discussed in Narendranath et al. [2023a]. So these unusual spikes in data are false positives. As discussed before Al and Mg do have a negative correlation. The same can be said for the Al-Fe, and Fe and Mg share a positive correlation. So both Fe and Mg have a higher presence in Mare basalt regions, largely in minerals like pyroxenes, olivines, and ilmenites. The detailed plots of the Fe/Si ratios on the lunar surface have been shown above.

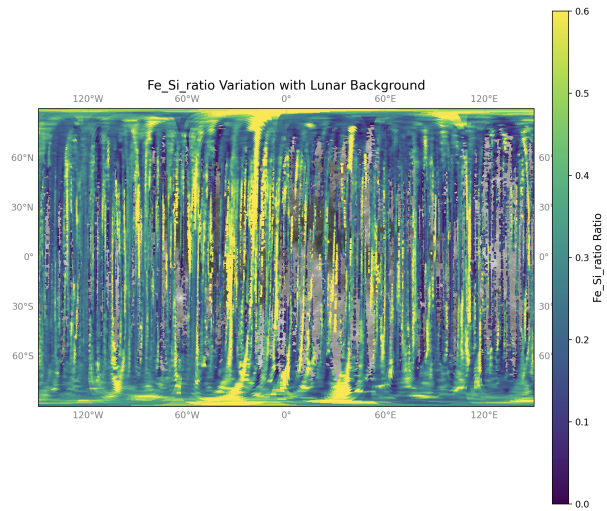


Fig. 20: Equidistant Cylindrical plots for Fe/Si ratios at 12.5km resolution, for nearside of moon. The mare regions (especially near Mare Imbrium and Oceanus Procellarum, Narendranath et al. [2023a]) have higher conc. Of Fe/Si ratios, with some occasional concentration in Highlands

The results show high Fe/Si concentration in Mare Basaltic regions as given in the literature. The highest concentrations (yellow grids) are all localized in and around Mare Imbrium. There are however certain high concentrations of Fe in Southern Highlands, which we mentioned earlier, which could be due to systematic uncertainties. Certain regions in the Highlands are seen to be completely depleted of Iron. All these results have a positive correlation with findings from Narendranath et al. [2023a].

As seen from the mean, the titanium has a pretty trace presence on the moon. The Ti also is difficult to detect due to it requiring high-intensity solar flares to produce XRF spectra. The graph shows a very skewed variation. The SD is also half of what the mean is, signifying, that most ratios fall in the ± 0.01 range. The majority of Ti lies in Mare basalt, with a very low presence in the Highlands. Basalts can be divided into High or Low Ti Basalts. In this graph, we can see a high density of possibly Low Ti Basalts, with few trailing, High Ti Basalts.

The plot for the Ti/Si ratio validates the literature data of high Ti presence in Mare Basalts

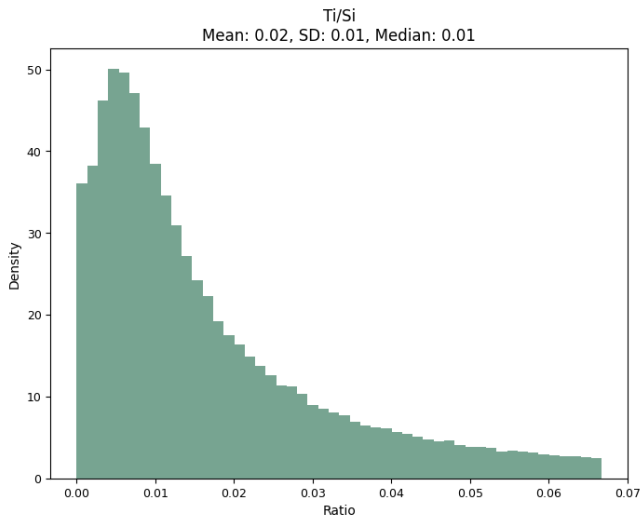


Fig. 21: Histogram plot showing normalized frequencies of Ti/Si ratio. The plot is heavily skewed towards the right, with higher ratios having longer tails.

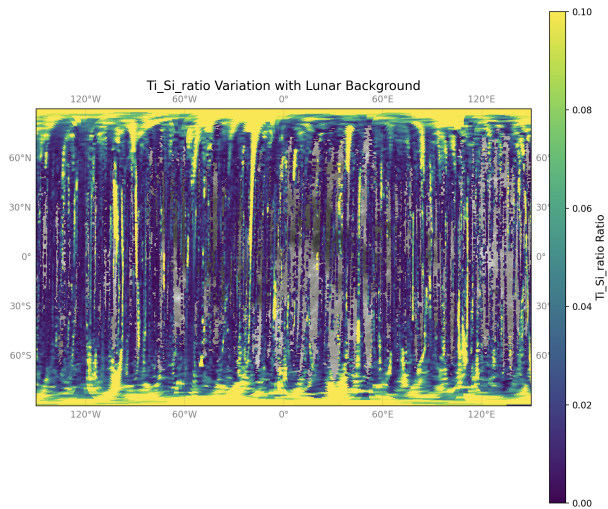


Fig. 22: Equidistant Cylindrical plots for Ti/Si ratios at 12.5km resolution, for nearside of moon. We can see mostly Ti devoid Highlands, with Low Ti Basalts dotted in between. We can see high Ti Basalts in Mare Imbrium

Haskin and Warren [1991]. We can see a very high concentration of Ti in Mare Imbrium which correlates with a similar high Fe presence in the same region. The high titanium basalt flows of mare Imbrium are the most iron-rich regions from the Moon which has been discussed in Narendranath et al. [2023a]. In contrast, the mare region south of the Copernicus crater is categorized as low titanium

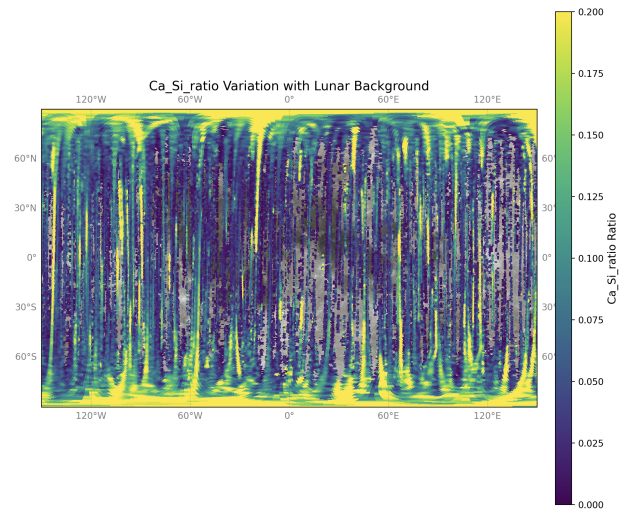


Fig. 23: Equidistant Cylindrical plots for Ca/Si ratios at 12.5km resolution, for nearside of moon.

basalts but with similar Fe abundances, which can be seen in the respective plots.

8.4. Calcium and Sodium

Calcium is an important component of lunar soil and is present majorly in the form of plagioclase feldspar in anorthositic rocks in Highland Terranes.

However, some calcium can still be found in mare basaltic regions, in the form of pyroxenes and other mafic materials Haskin and Warren [1991]. Ca resisted the harsh and volatile lunar environment as a refractory element, unlike Na. The Ca has been difficult to plot using XRF analysis due to difficulty in its excitation to flare events, hence limiting its coverage Narendranath et al. [2023a]. The plot for Ca shows limited coverage on the lunar surface, with very low presence in Mares, and validates good presence in southern Highlands and Oceanus Procellarum Narendranath et al. [2023a]. There's also an unusually high Ca presence near the Copernicus crater and Mare Imbrium. The impact events can also cause calcium-rich anorthositic rocks to get mixed with basalts and regolith (JOY et al. [2010]). This along with the presence of some

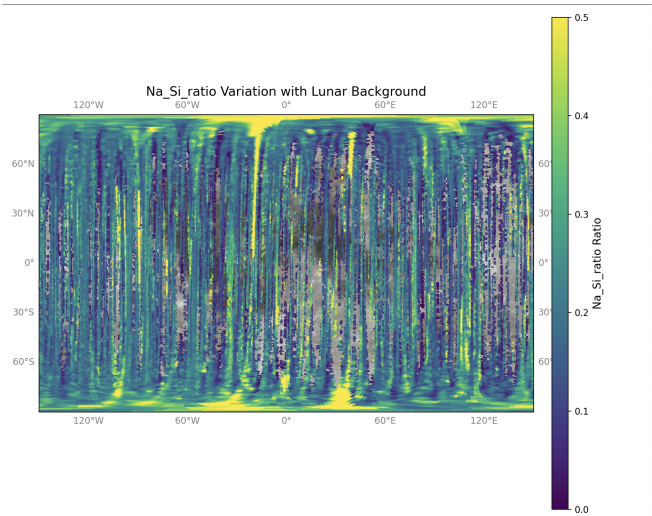


Fig. 24: Equidistant Cylindrical plots for Na/Si ratios at 12.5km resolution, for nearside of moon. The plot shows a near uniform yet low Na presence, with a high presence in Highland Terranes.

basaltic pyroxenes could explain the presence of high Ca presence.

Sodium in contrast to calcium, is a very volatile element, and most of the sodium has been lost since the high-temperature formation of the moon and due to the harsh lunar environment. The remaining sodium has been retained since, in mostly Highland Terranes in the form of minerals like the albite form of plagioclase feldspar, Haskin and Warren [1991]. The Na also helps in deducing the maturity of soil regolith, with older regoliths suffering significant Na depletion. The histograms for Ca and Na are provided in the appendix.

8.5. Manganese

The Mn is a very trace element as seen in the histogram (see appendix). The manganese generally doesn't follow much of any compositional heterogeneity as discussed in Haskin and Warren [1991] and typically correlates with Fe or Mg-rich sites. Most of Mn is present in diffuse regolith which has undergone various transformations in lunar crust evolution.

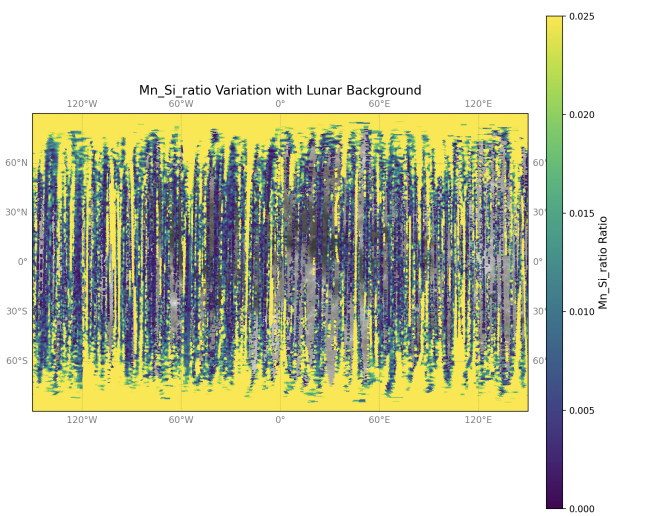


Fig. 25: Equidistant Cylindrical plot for Mn/Si ratios at 12.5 km resolution, for nearside of moon. We can see a loose trend, of Mn concentrating in loose regolith regions or Fe or Mg-rich sites

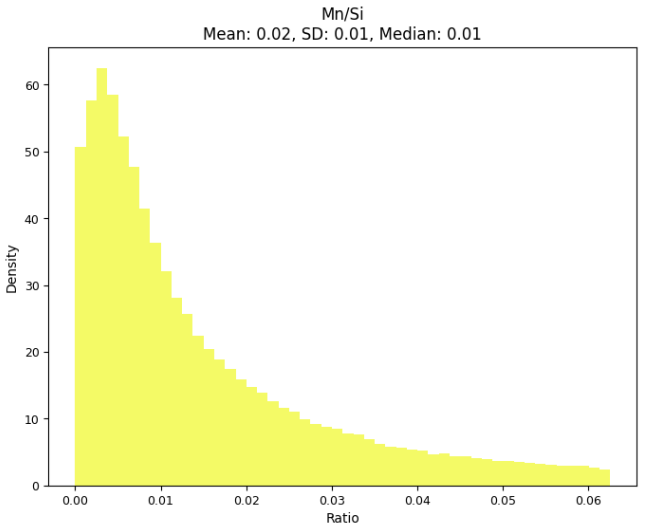


Fig. 26: Histogram plot for Mn/Si ratios. The graph is skewed similarly to that we got for Ti/Si

Ratio	Mean and Uncertainty
Mg/Si	0.81 ±0.21
Al/Si	1.39 ±0.3
Fe/Si	0.28 ±0.14
Ca/Si	0.05 ±0.04
Mn/Si	0.02 ±0.01
Na/Si	0.19 ±0.09
Ti/Si	0.02 ±0.01

Table 1. Uncertainties for the silicon reference ratios, derived from the standard deviations

9. Case Of Oxygen

Oxygen is available on the lunar surface in almost uniform quantity, but it is difficult to measure abundance using the XRF methods. It is because oxygen emits x-rays of very low energy, which is almost 0.52KeV which is below the detection limit of XRF equipment of CLASS instrument. The CLASS instrument is capable of measuring from an energy range of 0.8 KeV to 15 KeV. Radhakrishna et al. [2020b] Therefore we use a Lunar Prospector Gamma Ray Spectrometer (LP- GRS) abundance values for O, Ca, and Ti (except for spectra where Ca and Ti lines are observed in CLASS data). Narendranath et al. [2023b]

9.0.1. Our method for oxygen

When we analyzed the data using line flux measurements from CLASS, the results were not very accurate. This is because, as we discussed earlier, oxygen gets excited at very low energy levels, around 0.52 keV. At such low energies, the data picks up a lot of noise, which makes it harder to get reliable results. This problem can also be seen clearly in the figure below.

To solve this issue, we used a different approach. Instead of relying on the noisy measurements, we calculated the amount of oxygen by using stoichiometry. This means we estimated the oxygen abundance based on the ratios of other elements that we had already studied, such as FeO, TiO₂, CaO, Fe₂O₃, MgO, and SiO₂. This method worked much better because it avoided the noise problem. The results from this approach matched well with the data from the LP-GRS, which confirmed that our calculations were accurate. This shows that using stoichiometry is a better way to determine oxygen abundance in such cases. The histogram illustrates the Al/O ratio on the lunar surface for two datasets: LP-GRS (blue) and CLASS (orange). The LP-GRS distribution is concentrated around 0.1, indicating regions with lower aluminum content relative to oxygen. In contrast, the CLASS dataset

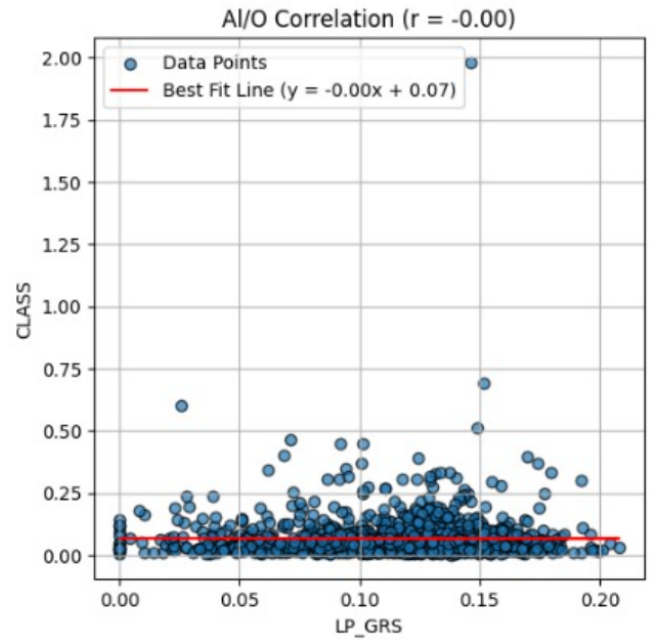


Fig. 27: This figure shows the data from CLASS did not correlate with the results from LP-GRS due to noises

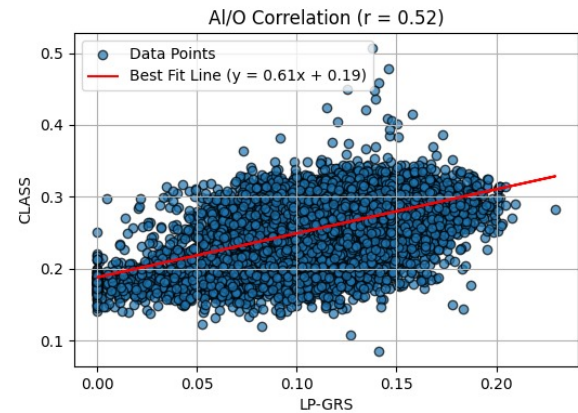


Fig. 28: This figure shows the correlation between Aluminium and oxygen

peaks around 0.2-0.3, suggesting areas with a higher concentration of aluminum, representing alumina-rich or feldspathic terrains.

9.1. Aluminium with oxygen

The density plot of Al/O ratios shows the distribution of aluminum abundance on the lunar surface. The curve peaks at approximately 0.27, indicating the presence of aluminum-rich feldspathic highlands. The narrow distribution

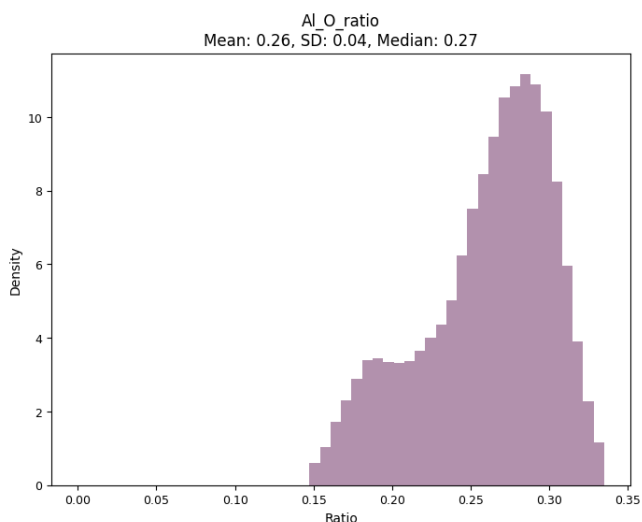


Fig. 29: Graph showing the density of Al/O ratios at 12.5 Km resolution.

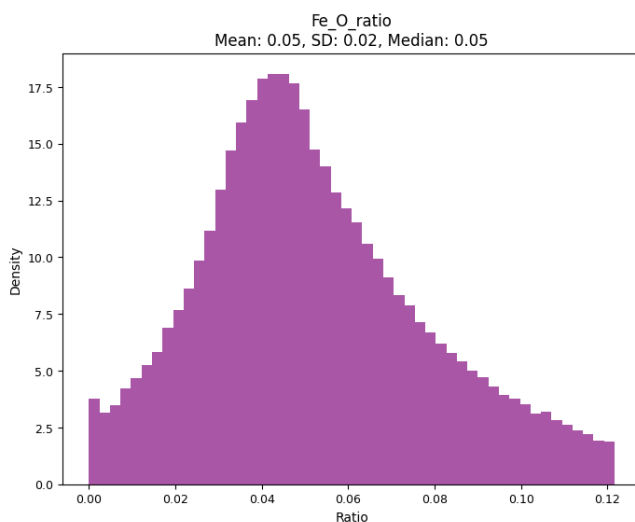


Fig. 31: Histogram showing the density of Fe/O ratios at 12.5 Km resolution.

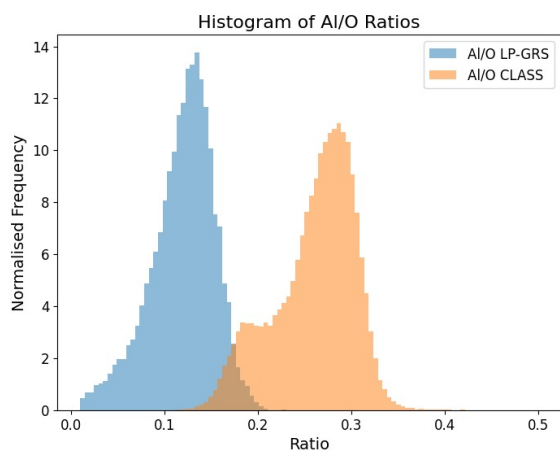


Fig. 30: This figure illustrates the correlation between data from CLASS and data from LP-GRS

with limited variability highlights compositional uniformity across the sampled regions. The Al/O ratios span 0.15–0.35, dominated by feldspathic regions, with minimal influence from mare basalts.

The graph depicts the correlation between the Al/O (aluminum to oxygen ratio) of CLASS and LP-GRS (Lunar Prospector Gamma-Ray Spectrometer) data, revealing a linear relationship with a moderate correlation coefficient of $r = 0.52$.

This positive correlation indicates that higher LP-GRS values typically correspond to higher Al/O ratios, essential for identifying feldspathic regions on the lunar surface. The bi-model is validated because of different concentrations in mare and highlands. The linear model, with a slope of 0.61 and an intercept of 0.19, shows that Al/O increases moderately with LP-GRS data. Low Al/O ratios are likely linked to iron and magnesium-rich mare basalts, while high Al/O ratios suggest the feldspathic highlands dominated by plagioclase feldspar. The distribution of aluminum, being more abundant than iron, aligns with minerals characteristic of the highlands, distinguishing them from mare regions.

9.2. Iron with oxygen

The histogram of Fe/O ratios on the lunar surface shows a slightly skewed distribution with a peak at lower values, indicating that iron is relatively scarce compared to oxygen, especially in the feldspathic highlands. Higher ratios correspond to iron-rich mare basalts. The quantitative analysis reveals a mean and median of 0.05 and an SD of 0.02, signifying a low and uniform iron presence. Lower Fe/O ratios mark iron-depleted highland regions,

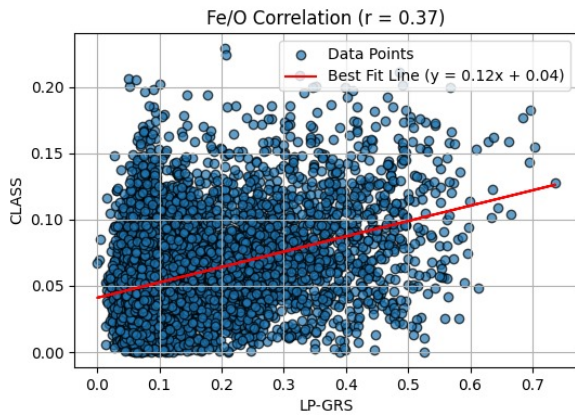


Fig. 32: This figure shows the correlation between Iron and oxygen

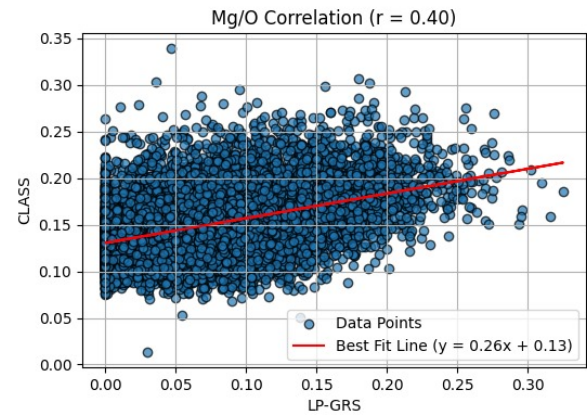


Fig. 34: This figure shows a correlation between Magnesium and oxygen

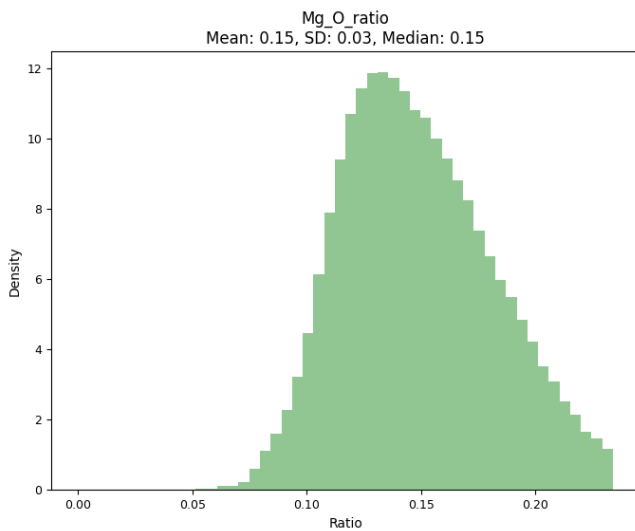


Fig. 33: Histogram showing the density of Mg/O ratios at 12.5 Km resolution.

while higher ratios point to iron-enriched mare areas. The low SD suggests that iron distribution is more consistent compared to other elements like magnesium and aluminum.

9.3. Magnesium with oxygen

The histogram of Mg/O ratios shows a bell-shaped distribution, indicating a near-normal distribution centered around the mean (~ 0.15). Most data points are clustered near this mean, with a lower density at the extremes. The quantitative analysis confirms this with a mean and median of 0.15 and an SD of 0.03, suggesting

low variability and symmetry. The range spans approximately 0.05 to 0.2, with peak density at the center. On the lunar surface, this central Mg/O ratio suggests the prevalence of magnesium-rich minerals, such as pyroxenes and olivines, common in mare basalt compositions. The low variability indicates a uniform distribution of magnesium content, distinguishing mafic regions with higher concentrations from feldspathic areas. The histogram comparing Mg/O ratios from LP-GRS (blue) and CLASS (orange) shows slightly right-skewed distributions, with CLASS peaking at higher values and broader variability, indicating more compositional differences. LP-GRS peaks at 0.08–0.1, while CLASS peaks at 0.12–0.15, with CLASS having a wider range up to 0.35 compared to 0.25 for LP-GRS. Both datasets overlap significantly, but CLASS data shows higher Mg content, suggesting detection sensitivity differences. Lower Mg/O ratios indicate feldspathic, magnesium-poor regions, while higher ratios point to mafic-rich lunar mare basalts.

10. Sub-Pixel Results and Methodology

The resolution enhancement process involves creating a uniform 5 km x 5 km grid over the lunar surface. Although this assumes a relatively flat surface of the individual cells, as we go to

higher resolutions from the original 12.5 km x 12.5 km, it becomes closer to the actual sphere thus reducing error. Each fit file data was assigned to one or more cells based on spatial overlap. For each grid cell, the elemental ratio was computed as a weighted average, where the weights are the percentage overlap of the raw data with the cell.

$$R_{weighted} = \frac{\sum_i (R_i \cdot A_{overlap,i})}{\sum_i A_{overlap,i}}$$

Aggregating raw data based on spatial overlap smoothens high-frequency noise, improving the reliability of calculated ratios for each grid cell. The use of weighted averages ensures that raw data points with significant spatial overlap contribute more heavily to cell values, maintaining fidelity to the underlying spatial patterns. (Fotheringham and Brunson [2010], Burt and Adelson [1983], L  uter [1988]).

In areas with sparse raw data, the calculated ratios are prone to higher uncertainty due to reduced data representation within the smaller grid cells. While the weighted averaging mitigates some of this error, the reliability of these values remains dependent on the density of raw data.

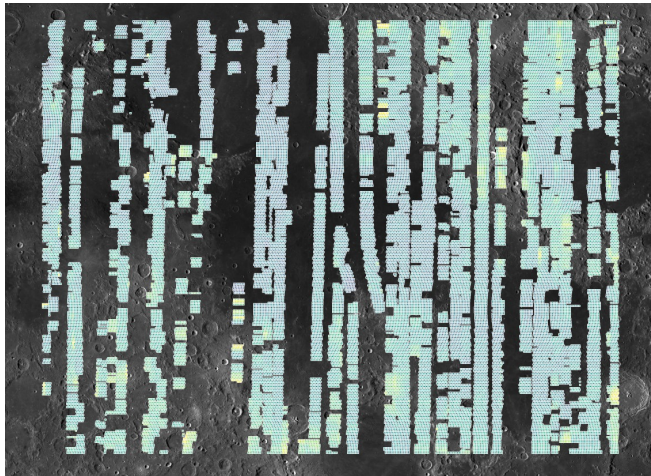


Fig. 35: CLASS data of Al/Si ratio on 5 Km mesh grid on Apollo 15 and 16 data site

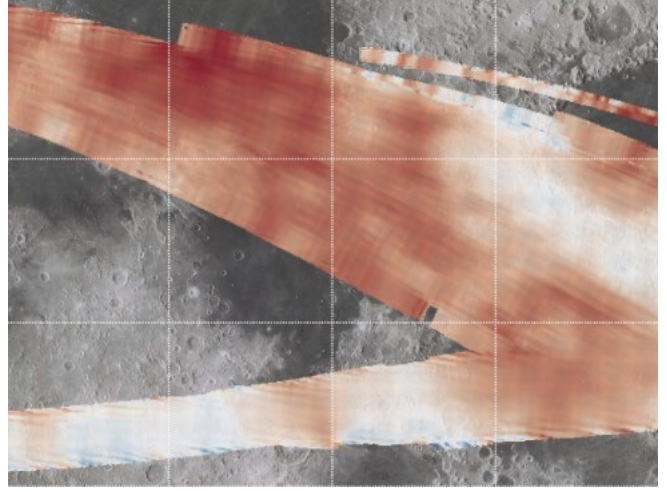


Fig. 36: Al/Si ratio on Apollo 15 and 16 data site Gloudemans, A. J. et al. [2021]

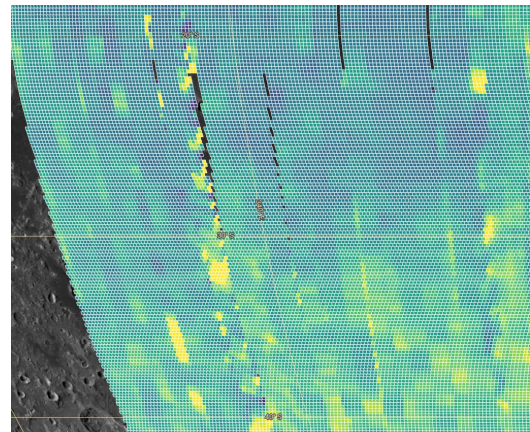


Fig. 37: Al/Si Ratio at a resolution of 5 km. Regions with higher concentrations of aluminum appear in distinct color patches, corresponding to lunar highlands (which are rich in feldspar), while lower ratios typically indicate basaltic regions found in lunar maria.

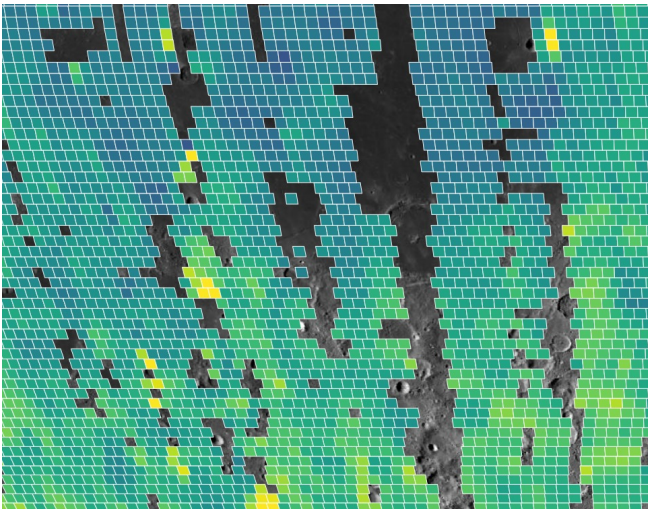


Fig. 38: Al/Si Ratio at a resolution of 12.5 km

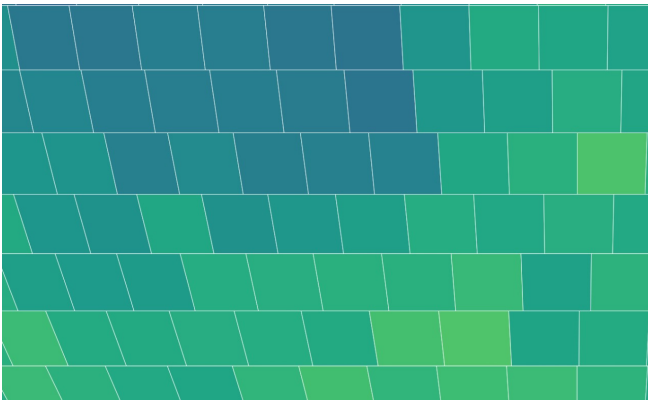


Fig. 39: Al/Si Ratio at a resolution of 100 km

We see a similar trend being followed in Fig. 33 and 34 with Apollo data and CLASS data showing similar high ratios in the region shown (dark red and blue color trend respectively). Gloudemans, A. J. et al. [2021]

The weighted spatial aggregation approach provides a reliable method for enhancing the resolution of elemental ratio datasets, particularly when the raw data is dense around a mesh grid. While it reduces random noise and retains broader spatial patterns, this method might not be a good way to go for regions of sparser raw data. Future improvements could involve adaptive grid sizing

or kernel-based weighting to better accommodate variations in data density.

11. Conclusion

This research effectively demonstrates the utility of X-ray fluorescence spectroscopy with enhanced data processing in mapping the elemental composition of the lunar surface. A novel spatial interpolation approach and noise-filtering algorithms were used to provide better resolution and accuracy than those of traditional temporal averaging methods. The use of mesh grid techniques to construct sub-pixel resolution maps helped make detailed maps of elemental distribution, especially the localized variations that had previously been undetectable.

Our analysis and subsequent plots provide us with detailed variations of Al, Mg, Mn, Ti, Ca, Fe, and Na on the lunar surface. The overall geographical results predict the high presence of Al and mild presence of Na and Ca in the Feldspathic Highland Terranes, with also little presence of Mg and Fe. It also predicts Mg-rich, Fe-rich and localized Ti-rich the Mare Basalts with little pockets of Na and Ca presence. The Mn appears to be diffuse in lunar regolith with little trend to follow.

These findings are compatible with existing models of the Moon and further confirmed with comparisons to the soil returned by the Apollo and Luna missions and by previous XRF instruments, including C1XS and LP-GRS. Moreover, the optimized line flux ratios instead of direct calculation of weight percentages were found efficient in reducing systematic uncertainties mainly when silicon is used as a reference element.

It addresses the difficulty of detecting elements like oxygen and titanium, whose detection requires other methodologies or solar flares with higher intensity to excite them. The stoichiometric method to estimate the abundance of oxygen offers a promising alternative when direct XRF measurements are noisy.

These results provide significant insights into the geological history of the Moon, its surface evolution, and compositional heterogeneity. The higher resolution and accuracy of elemental maps are critical for identifying potential sites for future lunar exploration, resource utilization, and establishing long-term human presence on the Moon. This research not only enhances our understanding of the lunar crust but also lays the groundwork for the development of more sophisticated techniques in planetary surface analysis for future missions.

12. Appendix

12.1. Explanation of Meshing

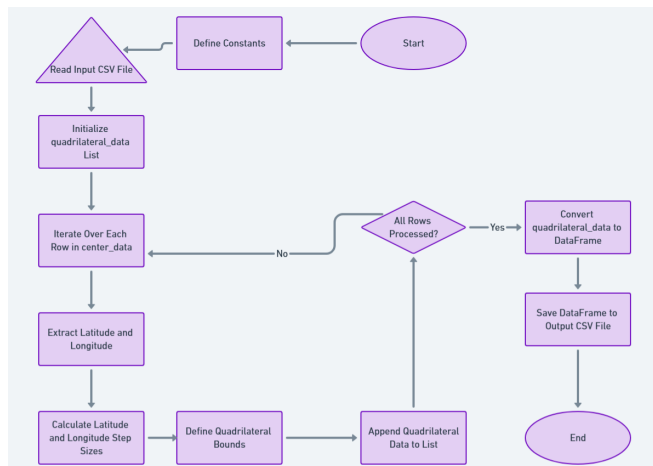


Fig. 40: Flow chart showing how to create mesh

12.2. Best Fit Ratios

We discussed exclusively the ratios taking Si as a reference as it has near uniform distribution on lunar surface. However, we present certain other ratios, which give us further insight into relative elemental variations. The Na/Mg ratio as shown has a high SD (50 percent of the mean), which could suggest a slight variation between their distribution. This makes sense as Na is a volatile element, and has a slightly higher presence in the Highlands compared to Basalts. Moreover, a mean of <1 , suggests that Na is less present than Mg,

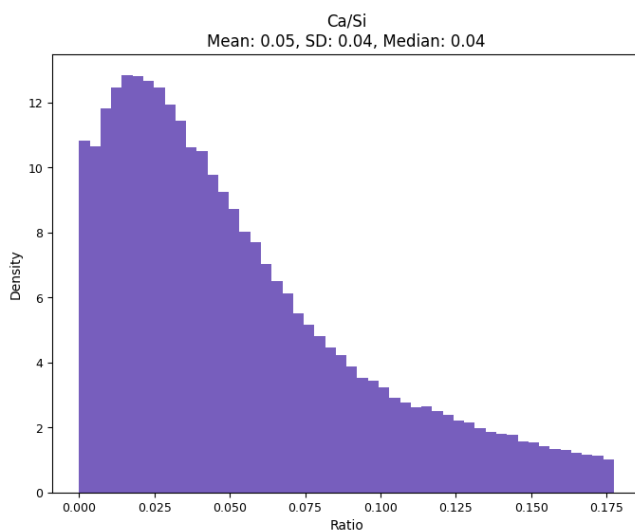


Fig. 41: Histogram plot of normalized frequencies of Ca/Si. The data is skewed towards the right, suggesting a more localized variation in ratios.

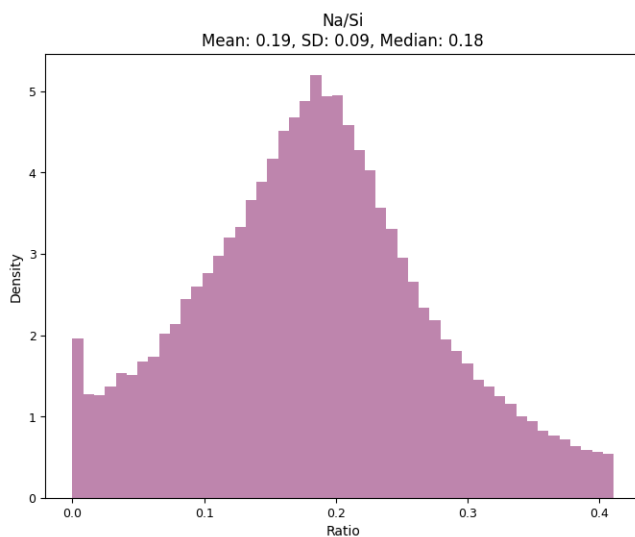


Fig. 42: Histogram plot of normalized frequencies of Na/Si. The data is broadly distributed, suggesting a more uniform variation on the lunar surface.

which also is correct Haskin and Warren [1991]. A similar argument can be made for Na/Fe ratios.

The Fe/Mg is another interesting ratio, which provides us insight into compositional heterogeneity. The histogram plot shows a high SD (50 percent of the mean). This again suggests a compositional heterogeneity between these elements. Although both Mg and Fe have a higher presence in Mares,

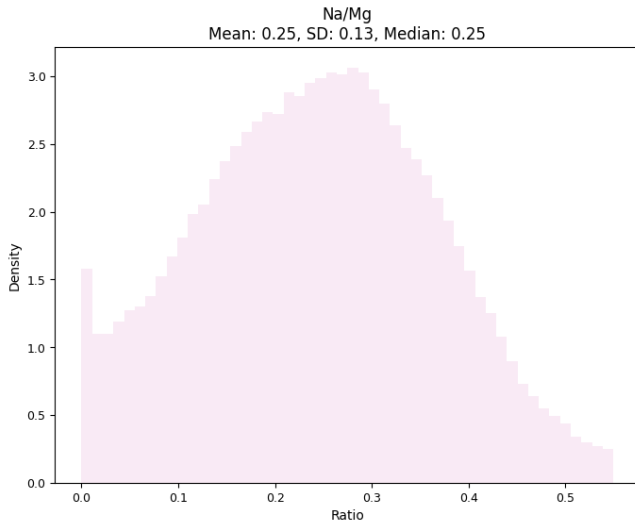


Fig. 43: Histogram plot of normalized frequencies Na/Mg ratios

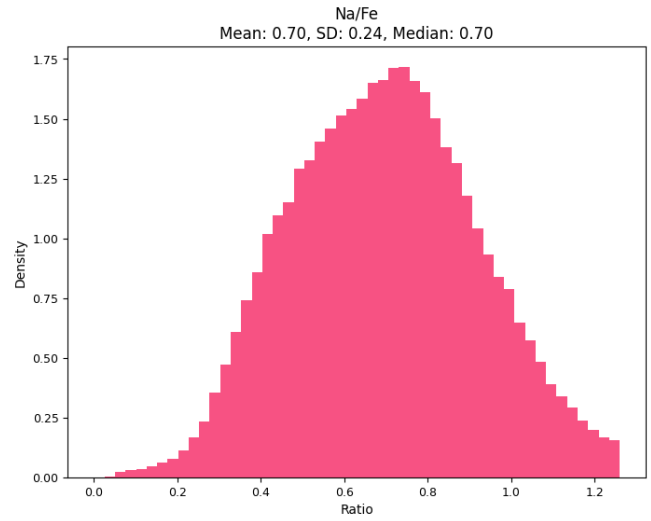


Fig. 44: Histogram plot of normalized frequencies for Na/Mg

Fe is more localized to Basalts whereas Mg has a uniform distribution, with a good presence in Highlands also. So Fe seems more localized compared to Mg, which has been discussed in Haskin and Warren [1991].

For Titanium ratios, we can see a near similar plot of Ti/Na and Ti/Fe with very high SDs ($>50\%$ of respective means). This again predicts a very localized variation of Ti compared to Na and Fe which is true, as most of Ti is locked in titanium-rich Mare Basalts, that too in localized Mares Narendranath et al. [2023a], Haskin and Warren [1991]. Moreover, we can see a higher relative SD of Ti/Na compared to Ti/Fe, suggesting a more localized distribution compared to Na than Fe. So this further points to slightly more diffuse sodium compared to Fe, Haskin and Warren [1991].

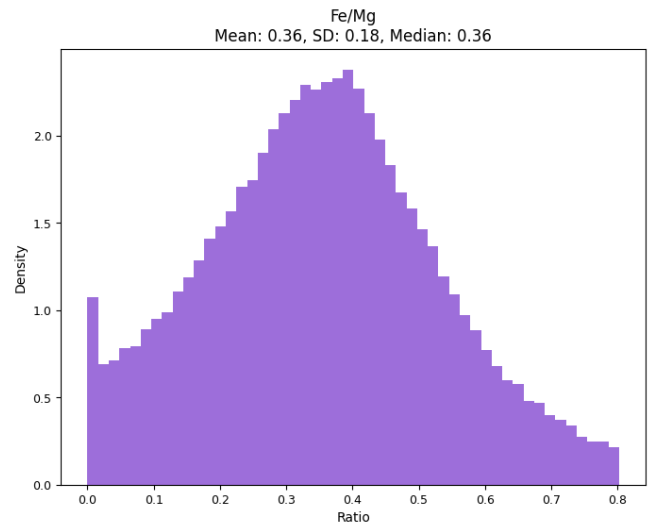


Fig. 45: Histogram plot of normalized frequencies for Fe/Mg

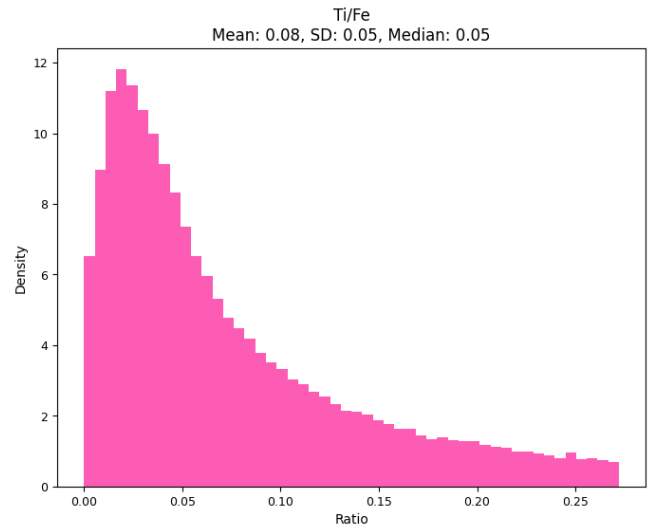


Fig. 46: Histogram plot of normalized frequencies for Ti/Fe ratios

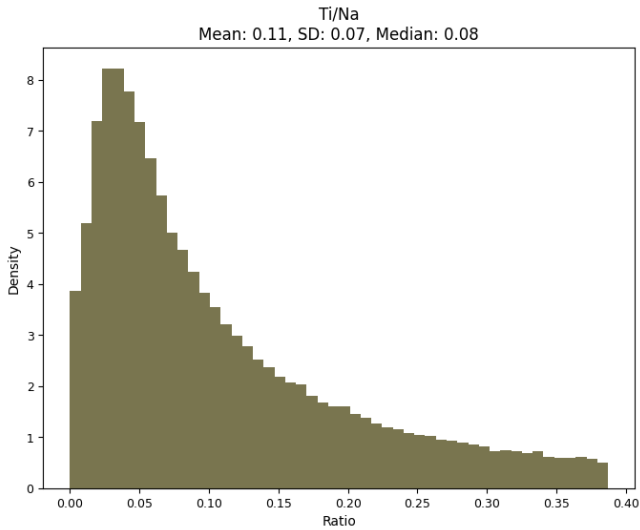


Fig. 47: Histogram plot of normalized frequencies for Ti/Na ratios

12.3. Algorithms Used in Methodology

Algorithm 1 Grid Processing Algorithm for Moon Data

```

1: Start
2: Load CSV data into a DataFrame
3: Generate Moon grid based on latitude and longitude
4: Build R-tree index for the grid
5: for each row in the CSV data do
6:   Validate coordinates
7:   if coordinates are valid then
8:     Create polygon for the quadrilateral
9:     if polygon is valid then
10:      Use R-tree to find potential grid cells
11:      Check intersections with grid cells
12:      if cells intersect then
13:        Update grid cell values: weighted sums, total
        weight, file count
14:      end if
15:    end if
16:  end if
17: end for
18: if all rows are processed then
19:   Calculate final weighted averages for each grid cell
20:   Save results to CSV
21: end if
22: End

```

13. Supplementary Data

The data used in the report can be found at:

- [The lunar soil compendium](#), used for validation and correlation with the XRF.
- [We used GOES data](#) for flare data
- [We used LP-GRS data](#) for oxygen

Algorithm 2 Processing FITS Files for Lunar Elemental Abundances

```

1: Start
2: Initialize SpectrumAnalyzer class
3: Read FITS files list from fits_list_csv
4: for each row in FITS list CSV do
5:   if FITS file exists then
6:     Open FITS file and read header/data
7:     if Solar angle  $\leq 90$  then
8:       Extract metadata from FITS header
9:       Compute counts and energy range from FITS data
10:      Fetch background data from CSV
11:    else
12:      Skip processing due to high solar angle
13:    end if
14:  else
15:    Log missing FITS file
16:  end if
17: end for
18: if all files processed then
19:   Write results to CSV
20: end if
21: End

```

Algorithm 3 Processing Directories and FITS Files for Solar Data Analysis

```

1: Start
2: for each directory in input_parent_dir do
3:   if directory exists then
4:     Process the directory
5:     for each .fits file in the directory do
6:       if file extension is .fits then
7:         Open .fits file and read header/data
8:         if Solar angle  $\leq 90$  then
9:           if Geotail check passes then
10:            Extract year, month, day from filename
11:            Update solar_angle_data list
12:            Accumulate counts and exposure
13:          end if
14:        else
15:          Skip file due to high solar angle
16:        end if
17:      else
18:        Skip non-.fits file
19:      end if
20:    end for
21:    if all files in directory processed then
22:      Normalize total counts by exposure
23:      Append accumulated data
24:    end if
25:  else
26:    Skip missing or inaccessible directory
27:  end if
28: end for
29: if all directories processed then
30:   Write solar_angle_data to CSV
31:   Write accumulated data to CSV
32: end if
33: End

```

Algorithm 4 Process Text Files and Extract Data by Date

```

1: Set folder_path
2: Initialize data_by_date as an empty dictionary
3: for each file in folder_path do
4:   if file does not end with ".txt" then
5:     Skip to the next file
6:   end if
7:   Extract date from the file name
8:   Open file and read line by line
9:   Initialize begin_end_times as an empty list
10:  for each line in file do
11:    if line starts with '' OR ':' then
12:      Skip to the next line
13:    end if
14:    Extract start_time, end_time, and flare_class from
line
15:    Validate start_time, end_time, and flare_class
16:    if validation fails then
17:      Skip to the next line
18:    end if
19:    Add {start_time, end_time, flare_class} to
begin_end_times
20:  end for
21:  Add begin_end_times to data_by_date[date]
22: end for
23: Write data_by_date to a CSV file with headers

```

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